Depropanizer Column Ver 1 · OTS

DC3-V1-PL-LAY-001

Plot plan and elevations

Rev 0 · Issued for teaching

Unit	Depropanizer Column Ver 1 · OTS
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Title	Plot plan and elevations
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Originator / DRAWN	Apinan Soottitantawat
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Note. *Numbers come from the simulator model (design case) and the assumptions listed; not for construction. Codes are named for the method only; no client, contractor, licensor or vendor is implied.*

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1 · Purpose and basis

Plot plan and elevations of the C-201 depropanizer unit, generated from the equipment list DC3-V1-ME-EQL-001, the datasheets DC3-V1-ME-DS-001, the line list DC3-V1-PE-LIN-001 and the EPC P&ID DC3-V1-PID-001 rev D. It fixes the position and height of every item, the roads, escape routes, crane access and the pipe rack, checks the layout against typical spacing rules, sets the drum elevations from the pump NPSH and the reboiler elevation from the sump level, and gives the reading level of every field instrument. The same geometry is written to layout.json for the Visual plant 3D tab of the OTS; the walking route of the outside operator is drawn on the plot plan.

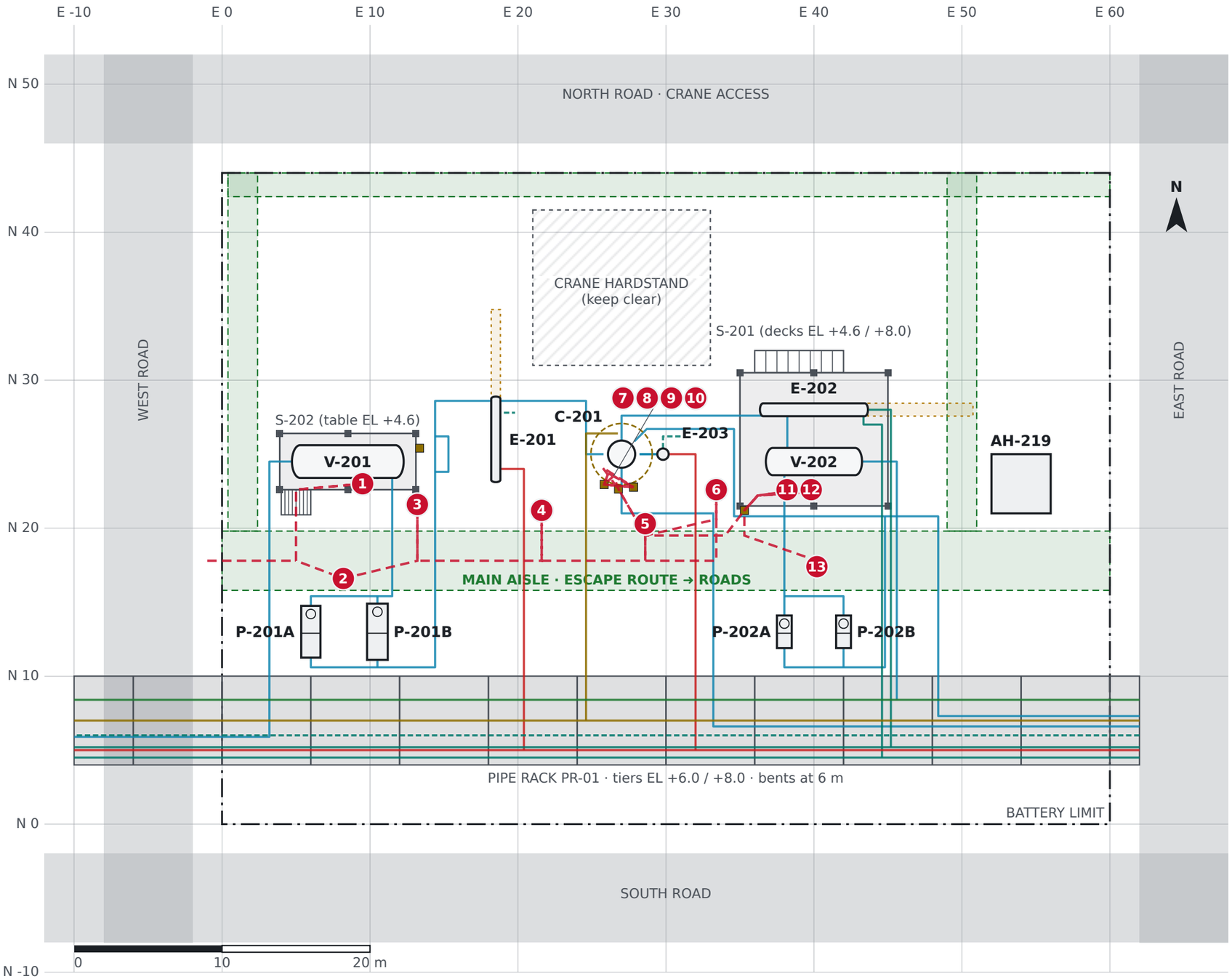
Layout basis (assumed)

- Coordinates: metres from the south-west corner of the battery limit, x east, y north; EL 0.000 = grade (high point of paving); plant north = true north (assumed).
- Prevailing wind from the south-west (bearing 225°, assumed); occupied buildings upwind.
- Spacing values are typical practice of the CCPS / insurer-table type, assumed for teaching; a project uses its own table and a siting study.
- Pump suction centreline EL +0.90; 0 % of the drum level ranges 0.15 m above the shell bottom; suction friction 2 kPa; NPSHr 3.0 m + 1.0 m margin (Process engineering basis); drum elevations rounded up to 0.1 m.
- Exchanger tubes: 19.05 mm × 4.88 m U-tubes (E-201), 19.05 mm × 6.1 m (E-202, split-ring floating head), 25.4 mm × 3 m (E-203 vertical thermosyphon); triangular pitch 1.25 do; top tubesheet of E-203 0.10 m below the sump NLL.
- Operator reach: reading points and hand wheels between a floor and 2 m above it; eye height 1.7 m; ladder flights ≤ 6 m; headroom over escape routes ≥ 2.2 m; rack tiers EL +6.0 / +8.0, bents at 6 m.
- Pumps: baseplates 1.3 × 3.5 m (P-201A), 1.4 × 3.8 m (P-201B turbine), 1.0 × 2.2 m (P-202A/B), assumed; control room, substation, analyzer house and flare locations assumed.

Sheets of this document

Sheet	Content
1	Plot plan 1 : 500 with key plan (control room, substation, flare direction), escape routes, crane hardstand, bundle-pull spaces and the operator round
2	C-201 elevation with platforms, ladders, nozzles, E-203 and the instrument reading levels
3	NPSH elevations of V-201 / P-201A/B and V-202 / E-202 / P-202A/B (sections looking west)

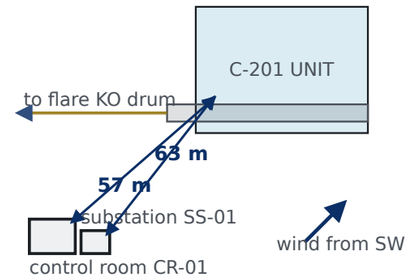
2 · Sheet 1 · Plot plan



Legend

- equipment (to scale)
- pipe rack PR-01
- structure / table top
- escape route / aisle
- crane hardstand
- bundle-pull space
- C-201 platforms (outline)
- ladder
- process line
- LP steam
- cooling water / condensate
- relief / flare
- fuel gas
- ① operator round station

Key plan (1 : 2000)



Prevailing wind from the south-west (assumed); occupied buildings upwind.

Coordinates in m from the south-west corner of the battery limit; EL 0.000 = grade. Scale 1 : 500 on A3.

Equipment to scale from the datasheets; pipe runs in plan; the dashed red line and numbered circles are the outside-operator round (section 7). Coordinates in m.

3 · Equipment locations and footprints

Tag	Item	Shape	x (E) m	y (N) m	Bottom EL	Top EL	Footprint m	Stands on	Basis
C-201	Depropanizer column	vertical cylinder	27.0	25.0	EL +0.00	EL +26.40	1.84 × 1.84	grade	ID 1.7 m, T/T 22.97 m, skirt 3 m (EQL / datasheet); OD incl. 18 mm wall + 50 mm insulation
V-201	Feed surge drum	horizontal cylinder (x-axis)	8.5	24.5	EL +5.20	EL +7.41	7.55 × 2.21	S-202	2.1 m ID × 6.5 m T/T; bottom EL from the P-201 NPSH (bubble-point basis)
V-202	Reflux drum	horizontal cylinder (x-axis)	40.0	24.5	EL +5.20	EL +7.04	6.50 × 1.84	S-201-1	1.8 m ID × 5.6 m T/T; bottom EL from the P-202 NPSH (bubble-point basis)
E-201	Feed preheater	horizontal cylinder (y-axis)	18.5	26.0	EL +1.19	EL +1.81	0.62 × 5.78	grade	BEU, shell ID 500 mm, 155 U-tubes 19.05 mm × 4.88 m (from 90 m²)
E-202	Overhead condenser	horizontal cylinder (x-axis)	40.0	28.0	EL +8.60	EL +9.47	7.30 × 0.87	S-201-2	AES, shell ID 850 mm, 795 tubes 19.05 mm × 6.1 m (from 290 m²)
E-203	Reboiler (vertical thermosyphon)	vertical cylinder	29.8	25.0	EL +0.88	EL +4.88	0.77 × 0.77	grade	BEM vertical, shell ID 650 mm, 305 tubes 25.4 mm × 3 m (from 73 m²); top tubesheet 0.10 m below the sump NLL
P-201A	Feed pump A (motor)	box	6.0	13.0	EL +0.00	EL +1.30	1.30 × 3.50	grade	BB multistage + motor; baseplate 1.3 × 3.5 m (assumed)
P-201B	Feed pump B (steam turbine)	box	10.5	13.0	EL +0.00	EL +1.40	1.40 × 3.80	grade	BB multistage + steam turbine; baseplate 1.4 × 3.8 m (assumed)
P-202A	Reflux / product pump A (motor)	box	38.0	13.0	EL +0.00	EL +1.10	1.00 × 2.20	grade	OH2 + motor; baseplate 1.0 × 2.2 m (assumed)
P-202B	Reflux / product pump B (motor)	box	42.0	13.0	EL +0.00	EL +1.10	1.00 × 2.20	grade	OH2 + motor; baseplate 1.0 × 2.2 m (assumed)
AH-219	Analyzer house (AT-219 / AT-220)	box	54.0	23.0	EL +0.00	EL +3.00	4.00 × 4.00	grade	from the Electrical tab hazardous-area table (AH-219/220); not on the P&ID
CR-01	Control room (off-plot, occupied)	box	−50.0	−36.0	EL +0.00	EL +4.50	16.00 × 12.00	grade	location assumed; non-blast-resistant
SS-01	Substation / MCC (off-plot)	box	−35.0	−38.0	EL +0.00	EL +5.00	10.00 × 8.00	grade	location assumed

Exchanger footprints from the datasheet areas

The datasheets give an area, not a size. The footprint follows from a tube choice (assumed: 19.05 mm tubes for E-201/E-202, 25.4 mm for the thermosyphon, triangular pitch 1.25 do): area per tube π·do·L, number of tube holes, then the bundle diameter Db = do·(Nt/K1)1/n1 and a shell clearance. For E-202: 290 m² ÷ (π × 0.01905 × 6.10) = 794.4 → 795 tubes; 2 passes (K1 0.249, n1 2.207): Db = 19.05 × (795/0.249)1/2.207 = 737 mm; split-ring floating head clearance 71 mm → shell ID 850 mm (rounded up to 50 mm). Overall length = tube length + channel and cover allowance (assumed).

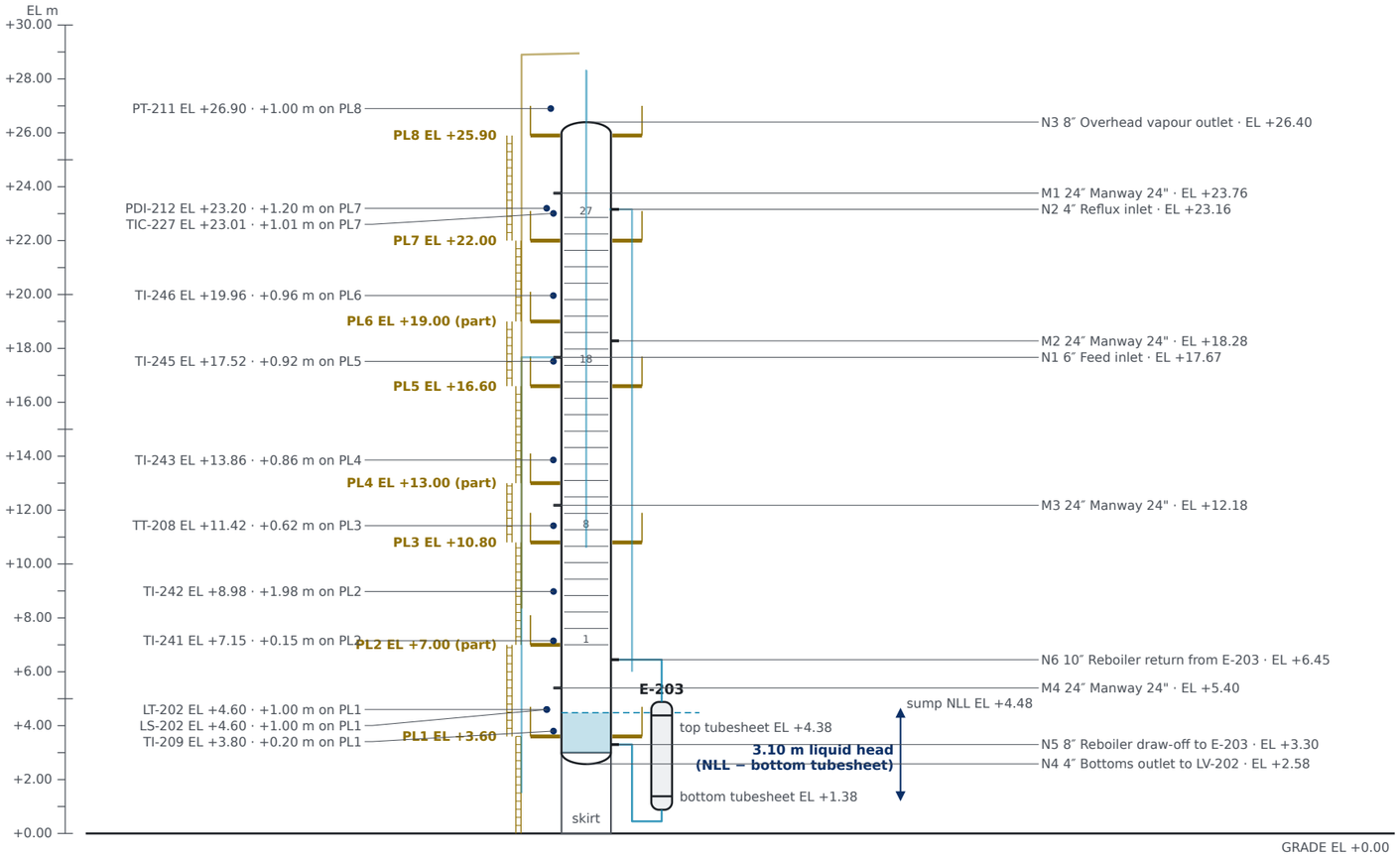
4 · Spacing and layout rules

Typical practice, assumed for teaching; every value computed from the layout geometry. 35 ✓, 0 ✗.

Rule	Typical practice (assumed)	Minimum	Pair / item	Value	Result
SR-01	Column and vessels, shell to shell (plan)	2 m	C-201 / V-201	13.81 m	✓
			C-201 / V-202	8.83 m	✓
			V-201 / V-202	24.48 m	✓
			C-201 / E-201	7.27 m	✓
			C-201 / E-202	8.78 m	✓
SR-02	LPG pumps to LPG vessels and the column	3 m	P-201A / V-201	8.64 m	✓
			P-201B / V-201	8.49 m	✓
			P-201A / C-201	21.87 m	✓
			P-202A / V-202	9.48 m	✓
			P-202B / V-202	9.48 m	✓
			P-202B / C-201	17.22 m	✓
SR-03	Pump to pump (maintenance access)	1 m	P-201A / P-201B	3.15 m	✓
			P-202A / P-202B	3.00 m	✓
SR-04	Thermosyphon reboiler close-coupled to its column (clear gap)	1 m	E-203 / C-201	1.50 m	✓
SR-05	Exchanger to other equipment	1.5 m	E-201 / V-201	5.92 m	✓
			E-201 / C-201	7.27 m	✓
			E-201 / P-201B	10.78 m	✓
			E-203 / V-202	6.56 m	✓
SR-06	Tube-bundle pull space: tube length + 1.0 m clear	1 m beyond the tubes	E-201 (pull N): clear 15.1 m to battery limit	15.1 m vs 5.88 m	✓
			E-202 (pull E): clear 16.4 m to battery limit	16.4 m vs 7.10 m	✓
SR-07	Equipment to the battery limit	3 m	nearest: AH-219	4.00 m	✓
SR-08	LPG equipment to the occupied control room (non-blast-resistant)	60 m	P-201A → control room CR-01	62.8 m	✓
SR-09	LPG equipment to the substation / MCC (ignition source)	30 m	P-201A → substation SS-01	57.4 m	✓
SR-10	Analyzer house to pump seals (outside the Zone 2 at grade)	7.5 m	P-202B seal → AH-219	12.4 m	✓
SR-11	Occupied buildings upwind: bearing to the control room within ± the angle of the wind-from direction	45 ° (max)	bearing unit → CR-01 234°, wind from 225°	9° apart	✓
SR-12	Main aisle clear width	3 m	ES-1 between the pump row and the equipment row	4.0 m	✓
SR-13	Headroom under pipes crossing escape routes	2.2 m	13 pipe crossings; lowest 4"-P-202 over ES-1	2.94 m	✓
SR-14	Pipe-rack lowest tier over roads	5 m	PR-01 tier EL +6.0 over the west road	6.0 m	✓
SR-15	Crane hardstand within this distance of the C-201 centreline, and clear	15 m (max)	HS-1 centre to C-201 11.3 m (edge 6.0 m); free of equipment and pipes	11.3 m	✓
SR-16	Travel distance from equipment to an escape route (plan)	25 m (max)	farthest: E-202	8.2 m	✓
SR-17	Escape routes from each elevated deck (stair or ladder)	2 routes	S-201 (S-201-1, S-201-2): 1 stair + 1 ladder route	2 routes	✓
			S-202 (S-202-1): 1 stair + 1 ladder route	2 routes	✓
			C-201 platforms: one caged ladder per flight to the platform below (column platforms, typical)	8 flights	✓
SR-18	Ladder flight between landings	6 m (max)	longest flight: S-201-LD1 (EL +0.00 → EL +4.60)	4.60 m	✓
SR-19	Relief line free-draining to the flare header, slope at least	0.002 m/m	6″-FL-601: 0 pocket(s); smallest fall 0.0167 m/m (1 : 60)	0.0167	✓

5 · Sheet 2 · C-201 elevation and platforms

C-201 · elevation looking north (true scale) with platforms, ladders, nozzles and reading levels



Platforms and decks

Platform / deck	On	EL	Kind	Extent	Serves
C201-PL1	C-201	3.60	ring full	r 0.97–2.07 m	sump manway M4, TI-209, LT-202 / LS-202
C201-PL2	C-201	7.00	ring part 135–315°	r 0.97–2.07 m	TI-241, TI-242, PDI-212 low tap
C201-PL3	C-201	10.80	ring full	r 0.97–2.07 m	manway M3, TT-208 (control tray)
C201-PL4	C-201	13.00	ring part 135–315°	r 0.97–2.07 m	TI-243
C201-PL5	C-201	16.60	ring full	r 0.97–2.07 m	manway M2, TI-245 (feed tray)
C201-PL6	C-201	19.00	ring part 135–315°	r 0.97–2.07 m	TI-246
C201-PL7	C-201	22.00	ring full	r 0.97–2.07 m	manway M1, TIC-227, PDI-212
C201-PL8	C-201	25.90	ring full	r 0.97–2.07 m	top: PT-211, vent, PSV-201 (davit)
S-201-1	S-201	4.60	deck	35–45 E, 21.5–30.5 N	V-202, LT-203 / LS-203, HV-216, FI-216
S-201-2	S-201	8.00	deck	35–45 E, 21.5–30.5 N	E-202, PV-210, HV-210, PT-210, TI-215
S-202-1	S-202	4.60	deck	3.9–13.1 E, 22.6–26.4 N	V-201, LT-201, LG-201, TI-201, drain
V201-TOP	V-201	7.46	deck	6.5–11.6 E, 24–25 N	V-201 top: vent, inlet nozzle

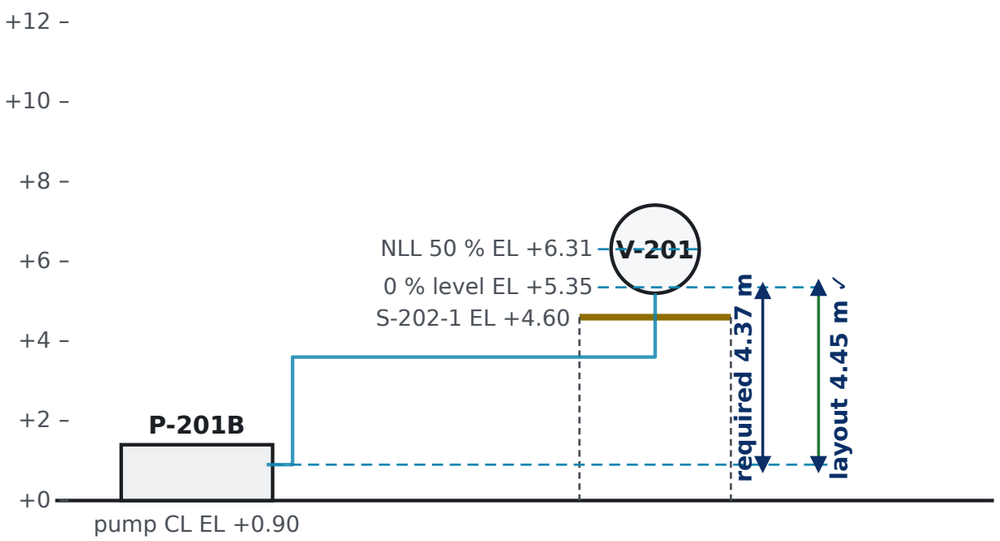
Ladders and stairs

Ladder / stair	On	From EL	To EL	Flight m	Note
C201-LD1	C-201	0.00	3.60	3.60	caged
C201-LD2	C-201	3.60	7.00	3.40	caged
C201-LD3	C-201	7.00	10.80	3.80	caged
C201-LD4	C-201	10.80	13.00	2.20	
C201-LD5	C-201	13.00	16.60	3.60	caged
C201-LD6	C-201	16.60	19.00	2.40	
C201-LD7	C-201	19.00	22.00	3.00	

Ladder / stair	On	From EL	To EL	Flight m	Note
C201-LD8	C-201	22.00	25.90	3.90	caged
S-201-LD1	S-201	0.00	4.60	4.60	caged · second escape (south-west corner), grade to the lower deck
S-201-LD2	S-201	4.60	8.00	3.40	caged · second escape, lower to upper deck (landing on the lower deck)
S-202-LD	S-202	0.00	4.60	4.60	caged · second escape (east end)
V201-LD	V-201	4.60	7.46	2.86	· to the V-201 top walkway
S-201-ST	S-201	0.00	8.00	—	stair · switch-back stair on the north face
S-202-ST	S-202	0.00	4.60	—	stair · stair from the main aisle

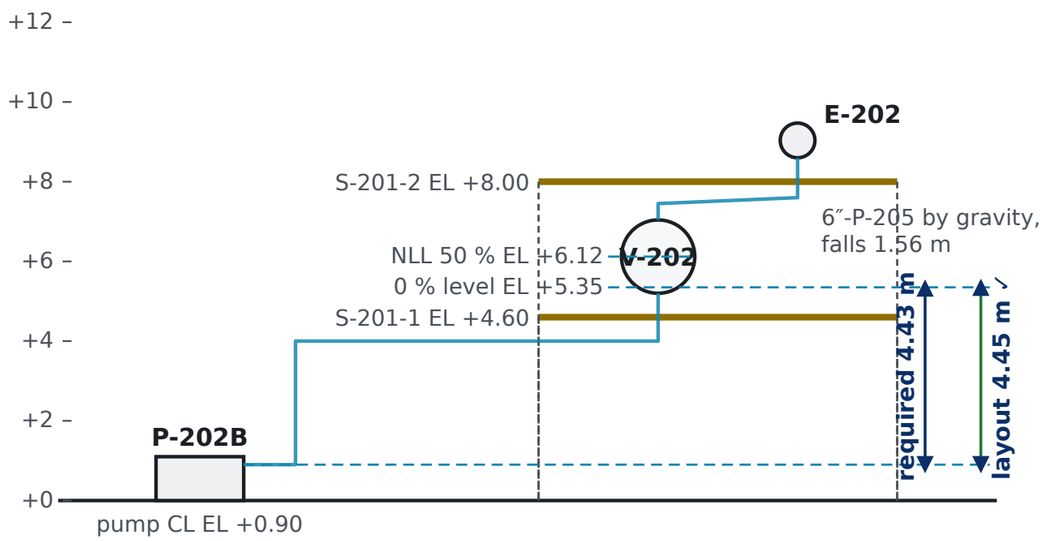
6 · Sheet 3 · NPSH elevations of V-201 and V-202, thermosyphon E-203

V-201 on S-202 and the feed pumps (section looking west)



required = NPSHr 3 + margin 1 + suction friction 0.37 m, liquid at its bubble point;
0 % of the level range 0.15 m above the shell bottom; drum bottom EL +5.20 (required EL +5.12)

V-202 / E-202 on S-201 and the reflux pumps (looking west)



required = NPSHr 3 + margin 1 + suction friction 0.43 m, liquid at its bubble point;
0 % of the level range 0.15 m above the shell bottom; drum bottom EL +5.20 (required EL +

NPSH of the pumps with the drum liquid at its bubble point

Drum → pumps	ρ kg/m³	Pressure above bubble, m liquid	hf m	Required static head m	Required 0 % level	Required shell bottom	Layout shell bottom		NPSHa at 0 % / NLL (m)	Static head at NLL
V-201 → P-201A/B	548.5	0.00	0.37	4.37	EL +5.27	EL +5.12	EL +5.20	✓	4.08 / 5.03	29.1 kPa
V-202 → P-202A/B	471.7	0.00	0.43	4.43	EL +5.33	EL +5.18	EL +5.20	✓	4.02 / 4.79	24.2 kPa

A centrifugal pump needs the liquid at its suction to stand above the vapour pressure by NPSHr plus a margin. Both drums hold liquid close to its bubble point: if the drum liquid is exactly at its bubble point, only the static head between the liquid and the pump is left, less the suction friction. The drum must therefore stand at least z0 % − zpump = NPSHr + margin + Δpf/(ρg) above the pump centreline, measured from the lowest level the drum is allowed to reach (0 % of the level range). For V-202: 3 + 1 + 2 × 1000/(471.7 × 9.81) = 4.43 m; with the pump centreline at EL +0.90 and the 0 % level 0.15 m above the shell bottom, the shell bottom must be at EL +5.18 or higher; the layout uses EL +5.20 (rounded up to 0.1 m). V-201 the same way gives EL +5.12 → EL +5.20. The model holds the drums above their bubble pressure (V-201 at 0.40 MPa g against a feed bubble pressure 312 kPa g; V-202 subcooled by the condenser), which is why the Process engineering tab finds NPSHa of 20.2 m and 40.5 m; the layout must not rely on it at start-up or after an upset.

Thermosyphon reboiler E-203

A thermosyphon circulates because the liquid leg from the sump to the bottom tubesheet is heavier than the boiling two-phase leg in the tubes. The usual arrangement puts the top tubesheet at, or just below, the normal liquid level of the sump; here the top tubesheet is at EL +4.38 (NLL EL +4.48 − 0.10 m, assumed), so the 3.0 m tubes put the bottom tubesheet at EL +1.38: a liquid head of 3.10 m (14.0 kPa at 460 kg/m³) drives the circulation. The return N6 enters at EL +6.45, 1.57 m above the E-203 top head, and above the 100 % level of LIC-202. The model does not compute the circulation (query EP-09); the elevation here is what a thermosyphon rating would start from.

7 · Instrument reading levels and hand-valve heights

Tag (P&ID)	Device	Kind	Where	EL	Floor	Height above floor	Reading / operating	
TI-201	TT-201	thermowell + transmitter	V-201	EL +5.70	S-202-1 EL +4.60	1.10 m	read	✓
LT-201		transmitter	V-201	EL +5.80	S-202-1 EL +4.60	1.20 m	read	✓
LG-201		gauge glass	V-201	EL +6.31	S-202-1 EL +4.60	1.71 m	read	✓
VENT-V201 rec.		hand valve	V-201	EL +7.71	V201-TOP EL +7.46	0.25 m	operate (NC)	✓
DR-V201 rec.		hand valve	V-201	EL +4.80	S-202-1 EL +4.60	0.20 m	operate (NC)	✓
MV-201		hand valve	6"-P-201	EL +1.30	grade	1.30 m	operate (NO)	✓
MV-202		hand valve	6"-P-201	EL +1.30	grade	1.30 m	operate (NO)	✓
NRV-201		check valve	4"-P-202	EL +1.90	grade	1.90 m	access for maintenance	
MV-203		hand valve	4"-P-202	EL +1.90	grade	1.90 m	operate (NO)	✓
NRV-202		check valve	4"-P-202	EL +1.90	grade	1.90 m	access for maintenance	
MV-204		hand valve	4"-P-202	EL +1.90	grade	1.90 m	operate (NO)	✓
XV-201		SIS valve	4"-P-202	EL +1.00	grade	1.00 m	access for maintenance	
FE-201		flow element	4"-P-202	EL +1.00	grade	1.00 m	access for maintenance	
FT-201		transmitter	4"-P-202	EL +1.30	grade	1.30 m	read	✓
MV-205		hand valve	4"-P-202	EL +1.00	grade	1.00 m	operate (NO)	✓
FV-201		control valve	4"-P-202	EL +1.00	grade	1.00 m	access for maintenance	
MV-206		hand valve	4"-P-202	EL +1.00	grade	1.00 m	operate (NO)	✓
MV-207		hand valve	4"-P-202	EL +1.00	grade	1.00 m	operate (NC)	✓
TIC-202	TT-202	thermowell + transmitter	6"-P-203	EL +1.50	grade	1.50 m	read	✓
HV-202		hand valve	6"-LS-301	EL +1.20	grade	1.20 m	operate (NO)	✓
XV-202		SIS valve	6"-LS-301	EL +1.20	grade	1.20 m	access for maintenance	
FE-206		flow element	6"-LS-301	EL +1.20	grade	1.20 m	access for maintenance	
FI-206	FT-206	transmitter	6"-LS-301	EL +1.30	grade	1.30 m	read	✓
TV-202		control valve	6"-LS-301	EL +1.20	grade	1.20 m	access for maintenance	
ST-201		steam trap	2"-CD-501	EL +0.45	grade	0.45 m	access for maintenance	
TI-209	TT-209	thermowell + transmitter	C-201	EL +3.80	C201-PL1 EL +3.60	0.20 m	read	✓
TI-241	TT-241	thermowell + transmitter	C-201	EL +7.15	C201-PL2 EL +7.00	0.15 m	read	✓
TI-242	TT-242	thermowell + transmitter	C-201	EL +8.98	C201-PL2 EL +7.00	1.98 m	read	✓
TT-208		thermowell + transmitter	C-201	EL +11.42	C201-PL3 EL +10.80	0.62 m	read	✓
TI-243	TT-243	thermowell + transmitter	C-201	EL +13.86	C201-PL4 EL +13.00	0.86 m	read	✓
TI-245	TT-245	thermowell + transmitter	C-201	EL +17.52	C201-PL5 EL +16.60	0.92 m	read	✓
TI-246	TT-246	thermowell + transmitter	C-201	EL +19.96	C201-PL6 EL +19.00	0.96 m	read	✓
TIC-227	TT-227	thermowell + transmitter	C-201	EL +23.01	C201-PL7 EL +22.00	1.01 m	read	✓
LT-202		transmitter	C-201	EL +4.60	C201-PL1 EL +3.60	1.00 m	read	✓
LS-202		transmitter	C-201	EL +4.60	C201-PL1 EL +3.60	1.00 m	read	✓
PDI-212	PDT-212	transmitter	C-201	EL +23.20	C201-PL7 EL +22.00	1.20 m	read	✓
PT-211		transmitter	8"-P-204	EL +26.90	C201-PL8 EL +25.90	1.00 m	read	✓
PSV-201		relief valve	8"-P-204	EL +28.75	C201-PL8 EL +25.90	2.85 m	access for maintenance	
DR-P210 rec.		hand valve	4"-P-210	EL +0.35	grade	0.35 m	operate (NC)	✓
LV-202		control valve	4"-P-210	EL +0.80	grade	0.80 m	access for maintenance	
FE-202		flow element	4"-P-210	EL +0.80	grade	0.80 m	access for maintenance	
FI-202	FT-202	transmitter	4"-P-210	EL +1.20	grade	1.20 m	read	✓
AI-220	AT-220	sample point	4"-P-210	EL +1.00	grade	1.00 m	read	✓
PI-205	PT-205	transmitter	E-203	EL +1.50	grade	1.50 m	read	✓
ST-202		steam trap	2"-CD-502	EL +0.50	grade	0.50 m	access for maintenance	
PI-207	PT-207	transmitter	6"-LS-302	EL +1.50	grade	1.50 m	read	✓
HV-209		hand valve	6"-LS-302	EL +1.20	grade	1.20 m	operate (NO)	✓
XV-209		SIS valve	6"-LS-302	EL +1.20	grade	1.20 m	access for maintenance	
TV-208		control valve	6"-LS-302	EL +1.20	grade	1.20 m	access for maintenance	
FE-205		flow element	6"-LS-302	EL +1.20	grade	1.20 m	access for maintenance	
FI-205	FT-205	transmitter	6"-LS-302	EL +1.30	grade	1.30 m	read	✓
PT-210		transmitter	8"-P-204	EL +9.20	S-201-2 EL +8.00	1.20 m	read	✓
TI-213	TT-213	thermowell + transmitter	6"-P-205	EL +7.40	S-201-1 EL +4.60	2.80 m	read	✗
PV-210		control valve	1"-P-206	EL +9.20	S-201-2 EL +8.00	1.20 m	access for maintenance	
HV-210		hand valve	1"-P-206	EL +9.20	S-201-2 EL +8.00	1.20 m	operate (NO)	✓
FE-217		flow element	1"-P-206	EL +9.20	S-201-2 EL +8.00	1.20 m	access for maintenance	
FI-217	FT-217	transmitter	1"-P-206	EL +9.10	S-201-2 EL +8.00	1.10 m	read	✓
LT-203		transmitter	V-202	EL +5.80	S-201-1 EL +4.60	1.20 m	read	✓
LS-203		transmitter	V-202	EL +5.80	S-201-1 EL +4.60	1.20 m	read	✓
HV-216		hand valve	10"-CW-401	EL +6.20	S-201-1 EL +4.60	1.60 m	operate (NO)	✓
FE-216		flow element	10"-CW-401	EL +6.20	S-201-1 EL +4.60	1.60 m	access for maintenance	
FI-216	FT-216	transmitter	10"-CW-401	EL +5.60	S-201-1 EL +4.60	1.00 m	read	✓
TI-214	TT-214	thermowell + transmitter	10"-CW-401	EL +6.20	S-201-1 EL +4.60	1.60 m	read	✓

Tag (P&ID)	Device	Kind	Where	EL	Floor	Height above floor	Reading / operating	
TI-215	TT-215	thermowell + transmitter	10"-CW-402	EL +9.90	S-201-2 EL +8.00	1.90 m	read	✓
MV-211		hand valve	6"-P-207	EL +1.30	grade	1.30 m	operate (NO)	✓
MV-212		hand valve	6"-P-207	EL +1.30	grade	1.30 m	operate (NO)	✓
NRV-211		check valve	4"-P-208	EL +1.90	grade	1.90 m	access for maintenance	
MV-213		hand valve	4"-P-208	EL +1.90	grade	1.90 m	operate (NO)	✓
NRV-212		check valve	4"-P-208	EL +1.90	grade	1.90 m	access for maintenance	
MV-214		hand valve	4"-P-208	EL +1.90	grade	1.90 m	operate (NO)	✓
PDT-230		transmitter	4"-P-208	EL +1.30	grade	1.30 m	read	✓
FE-204		flow element	4"-P-209	EL +1.20	grade	1.20 m	access for maintenance	
FT-204		transmitter	4"-P-209	EL +1.30	grade	1.30 m	read	✓
FV-204		control valve	4"-P-209	EL +1.20	grade	1.20 m	access for maintenance	
LV-203		control valve	3"-P-213	EL +1.20	grade	1.20 m	access for maintenance	
FE-203		flow element	3"-P-213	EL +1.20	grade	1.20 m	access for maintenance	
FI-203	FT-203	transmitter	3"-P-213	EL +1.30	grade	1.30 m	read	✓
AI-219	AT-219	sample point	3"-P-213	EL +1.40	grade	1.40 m	read	✓

8 · Outside-operator walking route

Thirteen stations, each a place an operator can stand (grade, a deck or a platform); camera points at 1.7 m eye height for the Visual plant 3D tab. The route is drawn on sheet 1.

No.	Station	Floor	Camera p (x, y, z)	Look at	Items
1	Feed surge drum V-201 (table top EL 4.6)	S-202-1 EL +4.60	9.5, 23.0, 6.3	9.8, 24.4, 6.1	V-201, TI-201, LT-201, LG-201, DR-V201, VENT-V201
2	Feed pumps P-201A/B	grade	8.2, 16.6, 1.7	8.3, 13.2, 1.4	P-201A, P-201B, MV-201, MV-202, NRV-201, NRV-202, MV-203, MV-204
3	Feed control station FV-201	grade	13.2, 21.6, 1.7	14.5, 24.1, 1.0	XV-201, FE-201, FT-201, MV-205, FV-201, MV-206, MV-207
4	Feed preheater E-201 and its LP steam	grade	21.6, 21.2, 1.7	20.0, 24.5, 1.2	E-201, HV-202, XV-202, FE-206, FI-206, TV-202, ST-201, TIC-202
5	C-201 base and the bottoms valve LV-202	grade	28.6, 20.3, 1.7	29.8, 21.7, 0.9	C-201, DR-P210, LV-202, FE-202, FI-202, AI-220
6	Reboiler E-203 and its LP steam	grade	33.4, 22.6, 1.7	31.5, 23.8, 1.3	E-203, HV-209, XV-209, TV-208, FE-205, FI-205, PI-207, PI-205, ST-202
7	C-201 platform EL +3.6 (sump level)	C201-PL1 EL +3.60	26.5, 23.5, 5.3	26.1, 24.0, 4.3	LT-202, LS-202, TI-209
8	C-201 platform EL +10.8 (control tray 8)	C201-PL3 EL +10.80	26.1, 23.7, 12.5	26.1, 24.3, 10.4	TT-208, TI-241, TI-242, TI-243
9	C-201 platform EL +16.6 (feed tray 18)	C201-PL5 EL +16.60	26.1, 23.7, 18.3	26.1, 24.2, 18.7	TI-245, TI-246
10	C-201 platform EL +22.0 and the top (overhead)	C201-PL7 EL +22.00	25.7, 24.1, 23.7	26.4, 25.3, 25.5	TIC-227, PDI-212, PT-211, PSV-201
11	Structure S-201 upper deck EL +8.0 (condenser and vent)	S-201-2 EL +8.00	39.0, 22.6, 9.7	42.4, 25.8, 9.3	E-202, PT-210, PV-210, HV-210, FE-217, FI-217, TI-215
12	Structure S-201 lower deck EL +4.6 (reflux drum)	S-201-1 EL +4.60	39.0, 22.4, 6.3	42.9, 25.0, 6.2	V-202, LT-203, LS-203, TI-213, HV-216, FE-216, FI-216, TI-214
13	Reflux pumps P-202A/B and the reflux / product station	grade	40.2, 17.4, 1.7	42.9, 16.7, 1.4	P-202A, P-202B, MV-211, MV-212, NRV-211, NRV-212, MV-213, MV-214, PDT-230, FE-204, FT-204, FV-204, LV-203, FE-203, FI-203, AI-219, AH-219

9 · Checks against the model and the other documents

41 ✓, 8 ✗. Each ✗ is explained in part 8.

Check	Layout value	Model / other document	Result	Note
Every equipment-list item is on the plot plan (P-201A/B and P-202A/B counted twice)	10 / 10	equipment list DC3-V1-ME-EQL-001	✓	
No two equipment bodies overlap; no body cuts another item's deck or platform (3-D boxes)	0 overlaps	13 bodies, 12 floors	✓	
Every route item exists in the layout (tag or equipment)	89 items, 13 stations	0 missing	✓	
Every field tag of the layout is on the EPC P&ID or marked as a recommended addition	78 tags (3 rec.)	DC3-V1-PID-001 rev D: 118 tags	✓	recommended: VENT-V201, DR-V201, DR-P210 (LA-06)
Every line number of the P&ID is routed in the layout	17 / 17	P&ID line numbers	✓	
ISO-1: line 6"-P-201 exists on the EPC P&ID	on the P&ID	DC3-V1-PID-001	✓	
ISO-1: 6"-P-201 size and design conditions = line list	6″, 1.35 MPa g / 65 °C	6″, 1.35 / 65 °C (DC3-V1-PE-LIN-001)	✓	
ISO-1: line 4"-P-202 exists on the EPC P&ID	on the P&ID	DC3-V1-PID-001	✓	
ISO-1: 4"-P-202 size and design conditions = line list	4″, 2.85 MPa g / 65 °C	4″, 2.85 / 65 °C (DC3-V1-PE-LIN-001)	✓	
ISO-1: line 6"-P-203 exists on the EPC P&ID	on the P&ID	DC3-V1-PID-001	✓	
ISO-1: 6"-P-203 size and design conditions = line list	6″, 2.85 MPa g / 175 °C	6″, 2.85 / 175 °C (DC3-V1-PE-LIN-001)	✓	
ISO-2: line 8"-P-204 exists on the EPC P&ID	on the P&ID	DC3-V1-PID-001	✓	
ISO-2: 8"-P-204 size and design conditions = line list	8″, 1.90 MPa g / 75 °C	8″, 1.90 / 75 °C (DC3-V1-PE-LIN-001)	✓	
ISO-3: line 4"-P-208 exists on the EPC P&ID	on the P&ID	DC3-V1-PID-001	✓	
ISO-3: 4"-P-208 size and design conditions = line list	4″, 2.60 MPa g / 70 °C	4″, 2.60 / 70 °C (DC3-V1-PE-LIN-001)	✓	
ISO-3: line 4"-P-209 exists on the EPC P&ID	on the P&ID	DC3-V1-PID-001	✓	
ISO-3: 4"-P-209 size and design conditions = line list	4″, 2.60 MPa g / 70 °C	4″, 2.60 / 70 °C (DC3-V1-PE-LIN-001)	✓	
ISO-4: line 4"-P-210 exists on the EPC P&ID	on the P&ID	DC3-V1-PID-001	✓	
ISO-4: 4"-P-210 size and design conditions = line list	4″, 1.95 MPa g / 140 °C	4″, 1.95 / 140 °C (DC3-V1-PE-LIN-001)	✓	
ISO-5: line 6"-FL-601 exists on the EPC P&ID	not numbered there	DC3-V1-PID-001	✗	LA-01 (number recommended)
ISO-6: line 6"-LS-302 exists on the EPC P&ID	on the P&ID	DC3-V1-PID-001	✓	
ISO-6: 6"-LS-302 size and design conditions = line list	6″, 0.70 MPa g / 175 °C	6″, 0.70 / 175 °C (DC3-V1-PE-LIN-001)	✓	
V-201: footprint length = T/T + two 2:1 heads from the equipment list	7.55 m	6.50 + 2 × 0.525 = 7.55 m	✓	
V-202: footprint length = T/T + two 2:1 heads from the equipment list	6.50 m	5.60 + 2 × 0.450 = 6.50 m	✓	
C-201: shell ID, T/T and skirt = datasheet	1.70 / 22.97 / 3.00 m	1.70 / 22.97 / 3.00 m	✓	
C-201 nozzle N1: the 6"-P-203 route ends at the datasheet elevation	EL +17.67	BTL EL +3.00 + 14.67 = EL +17.67	✓	
C-201 nozzle N2: the 4"-P-209 route ends at the datasheet elevation	EL +23.16	BTL EL +3.00 + 20.16 = EL +23.16	✓	
C-201 nozzle N5: the 8"-P-211 route ends at the datasheet elevation	EL +3.30	BTL EL +3.00 + 0.30 = EL +3.30	✓	
C-201 nozzle N6: the 10"-P-212 route ends at the datasheet elevation	EL +6.45	BTL EL +3.00 + 3.45 = EL +6.45	✓	
V-201: shell bottom high enough for NPSH with the liquid at its bubble point	EL +5.20	required EL +5.12 (NPSHr 3 + 1 m + hf 0.37 m)	✓	
V-202: shell bottom high enough for NPSH with the liquid at its bubble point	EL +5.20	required EL +5.18 (NPSHr 3 + 1 m + hf 0.43 m)	✓	
P-201 suction static head in the model (K.stat1) = layout head at the V-201 NLL	23.0 kPa	29.1 kPa (5.41 m × 548.5 kg/m³)	✗	LA-03 (± 10 %)
P-202 suction static head of the pump hydraulics = layout head at the V-202 NLL	30.0 kPa	24.2 kPa	✗	LA-03 (± 10 %)
E-203 top tubesheet at or below the sump NLL (thermosyphon)	EL +4.38	NLL EL +4.48	✓	
Reboiler return N6 above the E-203 top head (outlet rises to the column)	EL +6.45	E-203 top EL +4.88	✓	
Relief basis elevation of the C-201 bottom tangent = layout	EL +3.00 (DC3-V1-PE-RLB-001, assumed)	EL +3.00	✓	
Relief basis elevation of the V-201 shell bottom = layout	EL +1.00 (DC3-V1-PE-RLB-001, assumed)	EL +5.20	✗	LA-02
Relief basis elevation of the V-202 shell bottom = layout	EL +1.50 (DC3-V1-PE-RLB-001, assumed)	EL +5.20	✗	LA-02
Relief basis elevation of the E-202 shell bottom = layout	EL +8.00 (DC3-V1-PE-RLB-001, assumed)	EL +8.60	✗	LA-02
6"-P-205: condensate to V-202 falls all the way (self-venting gravity line)	0 rise(s) along the flow	0	✓	
6"-P-201: P-201 suction: no high point between V-201 and the pumps	0 rise(s) along the flow	0	✓	
6"-P-207: P-202 suction: no high point between V-202 and the pumps	0 rise(s) along the flow	0	✓	
6"-FL-601: relief tail pipe free-draining to the flare header	0 rise(s) along the flow	0	✓	
Headroom under every pipe crossing an escape route ≥ 2.2 m (assumed)	2.94 m (4"-P-202)	13 crossings	✓	
Every field reading point and hand wheel within 2.0 m of grade or a floor	52 / 53	TI-213 2.80 m	✗	LA-05
The four field valves of the OTS (FIELDV) are placed, every hand valve has a model key and normal position	4 / 4; 18 hand valves	FIELDV: HV216, HV209, HV202, HV210	✓	
Each C-201 manway has a full platform 0.3–1.9 m below it	4 / 4	M1 EL +23.76, M2 EL +18.28, M3 EL +12.18, M4 EL +5.40	✓	
PSV-201 tail-pipe equivalent length from the layout ≤ the relief-basis assumption	54.1 m (41.8 m + 4 elbows × 20 D)	30 m (DC3-V1-PE-RLB-001, ER-02)	✗	LA-07
layout.json validates (the same checks as validate.js)	valid	placed, overlaps, route, tags, lines	✓	

10 · Engineering queries

No.	Subject	Query	Proposed answer	Status
LA-01	Relief line from PSV-201 to the flare header not numbered	The EPC P&ID shows the PSV-201 outlet only as "4"x6"" to flare; the layout routes it as a 41.8 m tail pipe from the top platform down the column to the flare header on the rack top tier, and isometric sheet 5 needs a line number.	Number it 6"-FL-601 (class A1L, design 0.50 MPa g / 140 °C, MDMT −46 °C, assumed) and add it to the line list; if ER-02 is answered with an 8" tail pipe, the number becomes 8"-FL-601.	OPEN
LA-02	Relief fire-case elevations differ from the layout	The relief basis assumed V-201 bottom at EL +1.0, V-202 bottom at EL +1.5 and E-202 bottom at EL +8.0. The layout puts both drums at EL +5.2 (NPSH with the liquid at its bubble point) and E-202 at EL +8.6. Both drums remain inside the 7.6 m fire zone but the wetted area below 7.6 m changes.	Re-run the fire cases of DC3-V1-PE-RLB-001 with the layout elevations (V-201 / V-202 wetted area up to EL +7.6); keep the drum elevations, which also answer EP-12 (NPSH of P-201 without relying on the V-201 pressure).	OPEN
LA-03	Model static head of the pump suctions differs from the layout	The simulator uses a P-201 suction static head K.stat1 = 23 kPa and the pump hydraulics 30 kPa for P-202. With the layout elevations the static head at the normal level is 29.1 kPa (V-201, 5.41 m) and 24.2 kPa (V-202, 5.22 m).	Update K.stat1 to 29 kPa and the P-202 value to 24 kPa at the next model revision (raises NPSHa by 1.1 m; no effect on the exercises), or record the difference in the model assumptions.	OPEN
LA-04	TI-201 drawn in the V-201 vapour space	On the EPC P&ID the TI-201 bubble is connected to the top of V-201. A thermowell in the vapour space reads the vapour temperature, not the feed liquid the pumps take; the instrument list puts TT-201 on 6"-P-201.	Place the TI-201 thermowell in the V-201 liquid space (layout: 0.5 m above the bottom on the south side, read from the S-202 table top) or on 6"-P-201, and correct the P&ID at the next revision.	OPEN
LA-05	TI-213 not within reach (reading-level check)	TI-213 at EL +7.40 is 2.80 m above S-201-1: the 6"-P-205 drop from E-202 to V-202 has no floor below it within 2.0 m.	Move the TI-213 thermowell to the V-202 outlet 6"-P-207 near the pumps (same subcooled liquid, read from grade), or add a local platform on the V-202 top; the DCS reading is not affected.	OPEN
LA-06	Vents and drains unnumbered on the P&ID	The V-201 top vent, the V-201 bottom drain and the 4"-P-210 low-point drain are drawn as "VENT" and "DR" without tag numbers; the layout and the 3-D model need a tag for each hand valve (VENT-V201, DR-V201, DR-P210 used here, marked recommended).	Give every vent and drain valve a number in the valve list (e.g. HV-2xx) and show it on the P&ID; keep them normally closed and plugged.	OPEN
LA-07	PSV-201 tail-pipe length longer than the relief-basis assumption	ER-02 computed the built-up back-pressure with a 30 m equivalent length. The routed tail pipe is 41.8 m with 4 elbows, about 54 m equivalent (20 D per elbow, assumed), so the built-up back-pressure will be higher than in ER-02.	Recompute ER-02 with the layout length; consider an 8" tail pipe from the valve outlet, a balanced-bellows valve, or mounting PSV-201 lower on the overhead line (it must stay free-draining to the header).	OPEN
LA-08	Building siting and flare location assumed	The occupied control room is placed 63 m and the substation 57 m from the nearest LPG item (P-201A, P-201A), upwind for a south-west prevailing wind; the flare knock-out drum and the flare are off-plot to the west. None of this is based on a study.	Carry out the building siting study (explosion overpressure and toxic / fire exposure, API RP 752 / 753 practice) and the flare radiation study (API 521) before freezing the plot plan; confirm the prevailing wind from site data.	OPEN
LA-09	C-201 erection lift and the crane hardstand	C-201 weighs about 23.2 t empty (EQL estimate) plus platforms, ladders, insulation and dressing; it would be lifted from hardstand HS-1, 11.3 m from the column centreline, to a height of 26.4 m.	Select the crane and tailing crane with the dressed weight, confirm the hardstand bearing capacity with civil, and check the lift radius; dress the column (platforms, insulation) at grade where possible.	OPEN